Physics-Informed Graph Neural Networks with Uncertainty Quantification ~~: A Pre-Registered Evaluation Protocol~~ for Conservative, Registry-Compatible Blue Carbon ~~Credit Pricing~~Crediting

Nahla Nabil Skaik
Faculty of Computer Studies
Arab Open University
Kingdom of Bahrain
Email: nahla.nabil.52@gmail.com

***Abstract*—~~Flat, rule-based~~ Mangrove carbon credits are discounted by a non-permanence ~~buffers discount every credit in a voluntary carbon market identically, regardless of~~ buffer that ignores how uncertain the underlying carbon-stock estimate ~~'s confidence. We test whether~~ is. In this study, a physics-informed graph neural network (GNN) ~~with conformalized quantile regression (CQR) can supply a per-site confidence signal that complements, rather than replaces, this buffer, on a 10,124-node spatial graph of the Ten Thousand Islands mangrove fringe, southwest Florida (Sentinel-2, Global Mangrove Watch, field-calibrated~~ and gradient-boosted trees are combined to map soil organic carbon ~~). Model selection (physics-loss weight,feature set,calibration method)was pre-registered on validation data and confirmed once on a held-out test block. A five-seed check found~~ over two southwest Florida mangrove regions (10,124 and 22,326 nodes of 150 m), and conformal prediction turns each estimate into a calibrated 90% interval that feeds the uncertainty deduction of the Verra VM0033 methodology. Spatial cross-validation selected an ensemble of the physics-informed ~~loss's point-accuracy advantage over a no-physics GNN did not replicate: on test, the no-physics GNN has the best R^2 of three models (-0.383 vs. -0.430 physics-informed and -0.450 linear). Its predictions do correlate most strongly with the imposed physical prior on held-out nodes(0.755 vs. 0.682/0.686), evidence the constraint transfers under spatial shift, not independent physical validation,~~ GNN and XGBoost (cross-validated $R^2 = 0.47$ and 0.55), which reached test $R^2 = 0.28$ and ~~a real but much smaller margin than an earlier single-seed pilot suggested. Global CQR calibration reached 91.1% /96.4% coverage at 90%/95% nominal; a spatially localized (Mondrian) variant undercovered more on test (83.4% , 92.4%), a genuine negative result consistent with spatial-conformal theory's local-density requirement. A pricing proof-of-concept built on the better-calibrated global intervals maps interval width to a multiplier ranging 0.50–1.50 across five test nodes, against Plan Vivo's fixed 0.800 flat-buffer multiplier. Point-prediction accuracy remains weak throughout; the contribution is the pre-registered protocol itself, which caught both a single-run estimate and a calibration refinement overstating their reliability, not validated point accuracy or a validated pricing model~~0.60 against 0.20 and 0.59 for XGBoost alone. Conformal intervals covered 91.5% and 89.1% of test nodes, with interval width tracking absolute error (Spearman $\rho = 0.54$ in the first region). The registry rule withheld a median 12.7% and 2.1% of credits per node, an amount comparable to the 11.0% median buffer that seven issuing blue carbon projects actually hold. Aggregating node uncertainty as the methodology prescribes, assuming independent errors, covered only 18% and 33% of realized project-level errors on validation data, while a correlation-aware aggregation covered at least 91%. Against 87 field soil cores, the ensemble's site-level median error was 10.1%, below a constant regional mean (15.0%) and two published carbon maps (14.1% and 15.1%). These results support crediting rules that keep estimation uncertainty separate from non-permanence risk and that aggregate it under spatially correlated errors.

***Index Terms*—blue carbon, graph neural ~~network, physics-informed learning,~~ networks, conformal prediction, ~~uncertainty quantification, carbon credit pricing~~ carbon credits, uncertainty quantification**

I. Introduction

Voluntary carbon markets discount mangrove credits for non-permanence risk (carbon later released ~~via~~ through erosion, sea-level rise ~~,~~ or disturbance) ~~using a buffer pool: a fixed percentage~~ by withholding a fixed share of issued credits ~~withheld from sale, unconditioned on how confident the underlying carbon-stock estimate is~~in a buffer pool [1], [2]. That share does not depend on how well the carbon stock itself was estimated. Verra's VM0033 ~~and~~ Plan Vivo V5.0 both apply this uniformly, regardless of ~~sub-project granularity~~ [1], [2], and existing methodology does contain a separate uncertainty deduction: credits are reduced when the 90% confidence interval exceeds 20% of the estimate [1, p. 64]. In practice, the carbon-stock ~~models for this ecosystem and region report only point estimates ~~, maps available for mangroves report point estimates with no calibrated ~~per-prediction confidence a pricing rule could condition on [?], [3], [4]. Conformal~~

prediction supplies valid, distribution-free uncertainty for earth-observation and spatial regression [?], [5], [6], but no existing work pairs that calibrated uncertainty with a physics-informed spatial predictor, couples it to a buffer-pool pricing mechanism, or tests whether a calibration strategy (global versus spatially localized) holds under one region's calibration-density constraints, the narrower gap this paper addresses (Section II surveys each literature in full). This paper combines a per-site interval [3], [4], and the methodology combines plot uncertainties under an independence assumption that spatial models rarely satisfy.

This paper asks whether calibrated, per-site uncertainty from a spatial model can supply that deduction in a way a registry could audit. A physics-informed graph neural network [?], [7], previously applied to mass-conserving flood forecasting [?], [8] but not, to our knowledge, blue carbon, with [7], [9] encodes neighborhood structure and a tidal-inundation prior; tree ensembles [10], [11] supply strong tabular baselines; and conformalized quantile regression [12] gives distribution-free uncertainty calibration to test whether estimation confidence can complement, not replace, a flat intervals. This study departs from earlier work in four ways: (1) the interval of every node is converted into the VM0033 deduction, kept separate from the non-permanence discount.

The scope is deliberately narrow: one pilot region, one physics constraint (a steady-state mass-balance proxy, not a full temporal flux equation), and one primary UQ method (CQR; deep ensembles and MC dropout are left for future work). Every model-selection decision, physics-loss weight, feature set, calibration method, was pre-registered on a validation split and confirmed once on a spatially distinct test block never used for any prior decision; we treat this two-stage protocol itself as a methodological contribution (Section III-C) , since a naive single-run estimate and a theoretically-motivated calibration refinement each turned out to mislead here (Section V) . What survives that scrutiny is smaller and more tightly bounded than either initial result suggested: a real but modest prior-generalization advantage for the physics-informed model, and evidence that localized (Mondrian) conformal calibration, plausible in theory and competitive on validation-internal evidence, underperforms the simpler global variant on this AOI's test data buffer and compared with the buffers held by issuing projects [13]; (2) the methodology's own aggregation rule is tested empirically and shown to understate project-level uncertainty when model errors are spatially correlated; (3) predictions and published maps are checked against 87 independent field soil cores [14] ; and (4) the analysis covers two regions, two image dates and leakage-free spatial splits, with every model choice fixed on validation data under rules committed before the experiments were run.

The scope is deliberately applied: the paper does not propose a new network architecture. The contribution is a crediting pipeline whose uncertainty is calibrated, auditable and conservative, together with evidence on where that uncertainty is and is not informative.

The contribution is this pre-registered evaluation and its honest limits, not a validated point-prediction accuracy claim or a validated pricing model.

II. Related Work

Physics-informed and graph learning. Physics-guided ML integrates scientific structure into learned learning adds scientific structure to data-driven models [15]; PINNs add a PDE-residual loss [7], Graph Network Simulators encode physics via local, permutation-invariant message passing [?] . The closest prior art, flood forecasting [?], [8], is entirely hydrological and does not target mangrove carbon or couple prediction to a priced instrument via UQ. physics-informed networks penalize violations of governing equations [7], and graph formulations have carried such constraints into flood forecasting [8]. GraphSAGE [9] learns from node neighborhoods inductively, which suits spatial grids split into separate regions.

The dominant approach is per-pixel tabular regression (random forest, XGBoost) **Mangrove carbon mapping.** Most mangrove carbon studies regress field or map values on spectral indices . Simard et al.'s global mangrove biomass product [?] was inaccessible here (NASA Earthdata login required). with random forests or boosted trees [16]. Global soil carbon products are themselves machine-learning outputs [4], [17], and regional maps such as Radabaugh et al. built a field-plot-calibrated, 30 m carbon-stock model for the southwest Florida coast, one sub-region matching our pilot AOI [3] [3] are calibrated on local field cores. Geomorphic setting and tidal regime control mangrove soil carbon [18], [19], and soil carbon maps can disagree substantially at fine scales [20]. Random cross-validation overstates accuracy on such data; spatial blocking is required [21], [22].

Sanderman et al. produced the global, field-calibrated 30 m SOC product [4] we train against. A 105-study review confirms belowground-carbon and valuation research remain under-studied versus aboveground biomass [16]. None uses a graph representation or a physical constraint, or reports a calibrated uncertainty interval (the gap this paper addresses).

Uncertainty quantification. MC dropout [?] is cheap but poorly calibrated under shift; deep ensembles [?] calibrate better but need multiple full models, as in climate-downscaling [?]. CQR [12] turns a point predictor into calibrated intervals with a coverage guarantee that holds without distributional assumptions, so long as calibration and test data are exchangeable, validated for earth observation [5] ; that guarantee is conditional on exchangeability spatial autocorrelation can violate. GeoConformal Prediction [6] addresses this with a spatially localized framework, motivating the Mondrian variant we test (Section III-F) Split conformal prediction gives finite-sample coverage under exchangeability [23], [24]; conformalized quantile regression adapts the interval width to each input [12] and has been applied to earth observation [5]. Spatial dependence weakens exchangeability, which motivates localized (Mondrian) calibration [6], [24], whose validity needs dense calibration data in each region

[25]. Interval quality is scored with the Winkler interval score [26], [27].

Crediting and markets. Blue carbon credits traded at a weighted average of US$26.96 per credit in 2022–2023 [28], and registry records report issuances, retirements and buffer deposits per project [13].

Research gap. The studies above either map carbon without calibrated per-site intervals, quantify uncertainty without connecting it to a crediting rule, or set buffers without reference to estimation error. No study, to our knowledge, (i) converts calibrated spatial intervals into a registry uncertainty deduction, (ii) tests the registry's aggregation rule against realized errors, and (iii) checks the result against field cores and market records at the same time.

III. Data and Methods

~~The pilot AOI is the~~

A. Regions, data and graph

Two southwest Florida mangrove regions were used: Ten Thousand Islands / Rookery Bay ~~mangrove fringe, southwest Florida (~~(R1; $-81.90°$ to $-81.40°$~~lon.~~, $25.90°$ to $26.30°$~~lat.) , chosen for free access to every needed layer: Sentinel-2 L2A via Microsoft Planetary Computer, Global Mangrove Watch (GMW)v3 extent [29], the~~) and the Everglades National Park coast (R2; $-81.20°$ to $-80.90°$, $25.30°$ to $25.60°$). The label is the 30 m soil organic carbon stock (0–100 cm, t C/ha) of Sanderman et al. ~~SOC product [4], and a NOAA CO-OPS tide gauge (station 8725114). Field-calibrated validation numbers came from Radabaugh et al. [3].~~

~~Native 30 m SOC grid cells were~~ [4]. Native cells are aggregated into 5×5 blocks (~~~~about~~ 150 m~~nodes) ; Sentinel-2 bands were resampled onto this grid (bilinear) and the GMW mask (nearest-neighbor), so the label was never resampled. A node was kept if its block-mean GMW mangrove fraction exceeded~~) kept where the Global Mangrove Watch 2020 fraction exceeds 0.5 ~~and it had a valid SOC value~~ [29], giving 10~~,~~,124 nodes ~~and (~~64~~,~~,366 directed ~~edges (8-connectivity)~~ 8-neighbor edges) in R1 and 22,326 nodes (143,034 edges) in R2. Node features ~~were NDVI, raw B04/B08/B11 reflectance,~~ are ten Sentinel-2 L2A bands and derived indices [30] for 6 February 2026, the same quantities for 15 August 2026 with cloud masking, elevation, and distance to the tide gauge~~; the label was mean SOC (~~t. While revising this work, the Sentinel-2 clip used in the earlier version was found to be shifted 9.18 ~~C/ha, 0–100~~km north of its true position, so that the mean normalized difference vegetation index (NDVI) of mangrove nodes was 0.003; re-applying the same shift to correctly referenced imagery reproduced the earlier features (correlation 0.91–0.97), and all features were rebuilt (mean NDVI 0.46).

Splits are spatial: fine cells of about 2.5 ~~cm). An initial contiguous-band split gave negative test R^2 for every model family, since the held-out band straddled a real regional carbon-density gradient (Radabaugh et al.'s sub-regional figures range 265.1 ± 43.2 to 409 ± 104.4 Mg C ha^{-1} [3]). We replaced this with an 8×8 checkerboard grid of spatial superblocks (~~~~km are assigned whole to training, validation or test (R1: 6 ~~km each, far larger than the 150 m node spacing) , assigned by cumulative node count rather than superblock count, since mangrove nodes concentrate in a thin coastal fringe.~~

~~An early version of this evaluation used one fixed split, directly comparing test-set metrics across model variants to decide the paper's central claim, using test data beyond its role as a single untouched confirmation. We corrected this before any experiment below: a fresh validation /test block was carved from superblocks never individually inspected for a decision (only touched via gradient updates); the previously-inspected split was folded into the new training pool, since training on previously-seen data does not leak, but evaluating on it again would. The corrected split is 6~~,~~,465~~train~~ /1~~,~~,583~~validation~~ /2~~,~~,076 ~~test nodes(label means 427.9/421.2/399.4 t/ha,AOI mean 421.0). Every later selection decision (physics-loss weight, feature set, calibration method) was made on validation only, by rules fixed beforehand, confirmed once on test (Section IV): a pre-registered "tested X, selected Y" claim this protocol makes defensible, surfacing a genuine instability a fixed split could not reveal~~nodes; R2: 14,499/3,351/4,476), and every model is trained and evaluated on subgraphs induced by one split, so no edge crosses a split boundary.

~~All models shared a 2-layer GraphSAGE backbone [9] (hidden dim.~~

B. Models and physics prior

Baselines are linear regression, a location-only random forest, random forest with and without neighbor-mean (spatial lag) features, and XGBoost [10], [11]. The graph model is a two-layer GraphSAGE network (hidden size 16, dropout 0.3~~), Adam(lr $= 0.01$, weight decay 10^{-3})~~, Adam, learning rate 0.01, early stopping on ~~validation MSE (patience 30). Baseline and~~ held-out groups). The physics-informed ~~variants differed only in loss function, isolating the physics term's effect. The full mass-balance relationship is $dC/dt =$ Accretion $-$ Export $-$ Decomposition; a single Sentinel-2 date and no flux-tower data ruled out estimating this directly. We imposed a~~ variant adds a steady-state prior~~instead: predicted stock was softly encouraged toward an equal-weighted, min-max-normalized (training data only) combination of an accretion proxy (NDVI) and an export proxy (tide-gauge distance, a positional stand-in for tidal energy absent a per-node hydrodynamic layer) . Both proxy components are also raw model features (Section III-B) ; withholding both from a no-physics baseline reduces its test prior-generalization correlation only modestly (0.682 to 0.654), so this is not narrowly a feature-overlap artifact. A second term penalized negative predictions, a mass-balance corollary, not independent.Both combined into one physics loss, added to the standard MSE data loss, with no fitted coefficients inside the proxy itself: $w_{\text{nonneg}} = 0.1$ was held fixed (never swept); λ was chosen by a pre-registered~~

sweep over $\{0.01, 0.05, 0.1, 0.3, 0.5, 1.0, 2.0\}$, five seeds each, on validation only, selecting the value with best mean validation prior-generalization correlation within 0.05 of the best mean validation R^2. This rule selected $\lambda = 0.1$ (mean val. $R^2 = -0.007$, mean val. prior-gen. corr. $= 0.670$), not the ad hoc $\lambda = 0.3$ from an earlier pilot pass, which the rule did not select (full 7-value table in the project's number-source log).

A quantile-head GraphSAGE (same backbone) was trained with pinball loss on the 2.5th, 5th, 95th, and 97.5th percentiles, data loss only (no physics loss; combining both was future work). Two feature sets, NDWI (McFeeters, Green/NIR) and NDMI (Gao, NIR/SWIR), were tested as additions to base, five seeds each, on validation only, under the same tolerance rule as λ; neither improved mean validation prior-gen. correlation within tolerance, so base was retained (Section IV-A).

Split-conformal calibration used the validation set: the conformity score $E_i = \max(q_{\text{lo}}(x_i) - y_i,\ y_i - q_{\text{hi}}(x_i))$ was computed there,

$$\mathcal{L} = \mathrm{MSE}(\hat{y}, y) + \lambda\,\mathrm{MSE}(\hat{y}_z, P_z) + w\,\overline{\mathrm{ReLU}(-\hat{y})^2}, \qquad (1)$$

where P is an equal-weight, coefficient-free combination of tidal inundation frequency (the share of 2025 hourly National Oceanic and Atmospheric Administration (NOAA) water levels above each node's U.S. Geological Survey (USGS) 3DEP lidar elevation, both in NAVD88) and red-edge canopy index (NDRE), reflecting the role of hydroperiod in sediment delivery and carbon preservation [18], [19]. The prior in Eq. 1 replaced the NDVI and gauge-distance proxy of the earlier version. The final candidate set added the unweighted mean of the physics-informed GNN and XGBoost (the ensemble), a stacked variant in which the GNN corrects out-of-fold XGBoost predictions, and a GNN trained on both image dates.

C. Uncertainty and crediting

Quantile models (GNN, physics-informed GNN, XGBoost), normalized conformal random forest and a quantile ensemble were calibrated by split conformal prediction on the $\lceil (n+1)(1-\alpha) \rceil / n$ quantile of $\{E_i\}$ adjusted the interval bounds (*global* calibration, one correction for every node). A *localized (Mondrian)* variant instead partitioned validation nodes by spatial superblock (Section III-B) , computing a separate correction per block ($\geq$validation split [12], [23], either globally or per spatial block (Mondrian, at least 30 points, else global fallback). Both, plus a density-weighted variant (Gaussian kernel, bandwidth = median nearest-neighbor distance) , were compared on a validation-internal holdout (val split 50/50 into calibration/check, never reading test ; Section IV-B). Localized (Mondrian) was carried forward to test; global was the comparator, density-weighted a non-adopted checkcalibration nodes per block). A constant predictor with conformal calibration is reported as the trivial interval. For node i with estimate $\hat{y}_i$ and 90% interval $[l_i, u_i]$, the relative uncertainty $U_i = (u_i - l_i)/(2\hat{y}_i)$ enters the VM0033 rule,

$$c_i = \hat{y}_i \left(1 - \max(0,\ U_i - 0.20)\right)(1 - b), \qquad (2)$$

where b is the registry's non-permanence buffer, applied unchanged. Estimation uncertainty and non-permanence therefore remain separate factors.

D. Evaluation protocol

Feature sets, model settings, the physics prior and the interval method were chosen by spatial five-fold cross-validation (CV) on training and validation nodes [21] , using selection rules written and committed before each experiment. The test split was read once for all pre-specified models. After that read, one further model (the ensemble) was pre-specified on cross-validation alone and evaluated in a second, declared test read; its test numbers are marked in Table I. Analyses that followed the first test read (generalization, two-date training, project aggregation, field cores, market data) used validation data or external data only and are reported as exploratory. Nine logged deviations and all rules are archived with the code.

IV. Experiments and Results

This section reports measurements only; their interpretation is deferred to Section V.

The three-way comparison at the earlier pilot's fixed weight ($\lambda = 0.3$, $w_{\text{nonneg}} = 0.1$) was re-run across five seeds $\{1, \ldots, 5\}$ on validation data.

A. Accuracy

Table I and Fig. 1(a) plots mean report accuracy. In R1 the ensemble reached test $R^2 = 0.277 \pm 0.035$ (mean $\pm$ one std. dev. for validation R^2 and prior-gen. correlation. Baseline GNN: R^2 mean $= 0.005$, std $= 0.019$; corr. mean $= 0.544$, std $= 0.056$. Physics-informed GNN: R^2 mean $= -0.040$, std $= 0.026$; corr. mean $= 0.632$, std $= 0.137$. Linear regression is deterministic: $R^2 = -0.030$, corr. $= 0.486$ (no error bar; no seed dependence to show). The λ sweep and NDWI/NDMI ablation (Section III-E/F) selected $\lambda = 0.1$ and the base feature set mechanically, by the rules stated where each is defined.

On the validation-internal calibration/check split (Section III-F), at 90 and 95% nominal respectively: global calibration reached 88.0 and 95.6% empirical coverage (mean width 307.6 and 367.8standard deviation (SD) over three seeds; mean absolute error (MAE) 58.1 t/ha) ; localized (Mondrian) reached 92.2 and 95.6% (width 305.3 and 358.2 tC/ha); density-weighted reached 81.1 and 84.7% (width 220.2 , against 0.203 for XGBoost and 0.170 for the physics-informed GNN, whose seed SD (0.054) was a quarter of the plain GNN's (0.236). A constant training mean scored -0.11, and the location-only model -0.79, although it led cross-validation in R1 (0.42). In R2 all non-linear models scored 0.56–0.60. On interior test nodes at least two hops from any other split, the ensemble kept $R^2 = 0.19$ (R1) and 241.5 t/ha) , narrower but visibly undercovering both levels.

TABLE I
MODEL ACCURACY (R^2) AND FIELD-SITE ERROR

Model	R1 CV	R1 test	R2 test	Field (%)
Constant (train mean)	–	−0.11	−0.01	15.0
Linear	0.22	−0.24	0.41	–
Forest on lon/lat only	0.42	−0.79	0.57	–
Random forest	0.38	0.03	0.58	–
XGBoost	0.41	0.20	0.59	16.5
GNN	0.39	0.07	0.58	–
Physics-informed GNN	0.41	0.17	0.56	11.6
Ensemble (GNN+XGBoost)	**0.47**	**0.28**†	**0.60**†	**10.1**

Test: mean of 3 seeds. †Second, declared test read. Field: median absolute error at 25 core sites; published maps 14.1 and 15.1.

Table ?? reports the single test evaluation (train 0.58 (R2). With all other settings fixed, correcting the image georeferencing alone raised the GNN's cross-validated R^2 : linear 0.236, baseline GNN 0.276, physics GNN 0.289; negative-prediction rate 0.0 for all three, not shown) . The train ranking (physics > baseline > linear)reversed on test: the no-physics baseline GNN had the best test $R^2$from 0.17 to 0.26. The physics prior's correlation with the predictions it constrains was 0.14 on R1 test nodes, against 0.76 for the earlier proxy.

Edge incoherence (neighboring test-node jumps exceeding the 95th pct. of real training-side neighbor differences, threshold 123.12 t/ha) was 0.0 for both GNNs, near-zero for linear (graph message-passing generically, not the physics term).

Table ?? reports both methods on the single test evaluation. Global met or exceeded nominal coverage at both levels; localized (Mondrian)

B. Generalization and stability

A single ensemble trained on both regions reached cross-validated $R^2 = 0.37$ (R1) and 0.53 (R2), pre-registered (Section III-F)as the method to carry forward, undercovered more at both levels, despite matching/exceeding global there (Section IV-B). One test node (of 2,076) got an inverted (empty) interval under block-local correction at both levels, floored to zero width and excluded from Section VI's pricing examples; either way it counted as a coverage miss. against 0.47 and 0.55 for region-specific models, whereas models applied to a region they had not seen scored below zero (Fig. 1(b)plots both against the $y = x$ perfect-calibration line.

An earlier single-seed pilot found). In that zero-shot setting, conformal intervals calibrated on the source region covered 76–89% of target validation nodes; recalibrating them on half of the target's validation nodes restored 88–97%. Training the physics-informed loss (Eq. ??) improved test GNN on both image dates raised the rank agreement of its predictions across dates from 0.44 to 0.88 and its R^2 over a no-physics baseline.The multi-seed check (Section IV-A) already showed this was fragile (physics-informed mean validation R^2 across 5 seeds (−0.040) sat below the baseline's (+0.005) even before the corrected split and pre-registered λ), and the final test evaluation confirms it: the no-physics baseline GNNhas the best test R^2 of the three. This is the actual finding, not reconciled with the earlier result by reframing; a single seed on a single split was not reliable here. with August imagery from −0.29 to 0.35, for a 0.02 loss with February imagery.

The physics-informed model's predictions correlate with the imposed mass-balance proxy more strongly than either non-physics model on every evaluation in this paper, multi-seed validation (0.632 vs. 0.544/0.486)and the single test evaluation (0.755

C. Intervals and crediting

Table II and Fig. 1(c) summarize the selected intervals (conformalized quantile GNN, global calibration). In R1 the intervals matched the trivial interval's width but scored 18% better on the Winkler score, and their width ranked absolute errors ($\rho = 0.54$, 95% confidence interval 0.51–0.58); in R2 they were 32% narrower than the trivial interval (191 vs. 0.682/0.686). This requires the same precise interpretation as before: the proxy correlated against is the exact quantity the physics loss trained toward, so this is a *prior-generalization* check (whether the constraint transfers to held-out nodes), not independent physical validation , which this AOI's single-date, flux-tower-free data cannot supply. Whether this correlation reflects genuine physical behavior, not just successful optimization against its own training target, is this paper's most significant open question. Resolving it would require flux-tower or repeated field-plot carbon-flux measurements at this AOI, which do not currently exist, the clearest concrete next step for extending this work beyond a proxy-based prior. What changed under the corrected protocol is the effect's size, not its direction: a real but modest margin on test (0.069282 t C/ha). Mondrian calibration covered 87.9% in R1, where only six validation blocks were available. Applied per node, the VM0033 rule withheld a median 12.7% (R1) and 2.1% (R2) of credits; across all five interval methods, credited stock stayed within 0.84–0.073) , not the 0.6-point gap the single-seed pilot reported. This is a direct product of the pre-registered protocol working as intended. .90 (R1) and 0.95–0.98 (R2) of the buffer-only amount. Interval width did not track 2010–2020 mangrove loss (Spearman −0.06 in both regions). Among the seven blue carbon projects with issued credits (6.62 million), the median buffer was 11.0% and 73.5% of credits had been retired [13].

Section III-F pre-registered localized (Mondrian)CQR from validation-internal evidence, before test was read. On test it undercovered more than global at both levels (83.4% vs. 90% nominal; 92.4% vs. 95%, against global's 91.1%/96.4%), likely because only 6 spatial superblocks had enough validation nodes (≥ 30) for a stable local correction. This matches theory: [25] shows that local conformal validity needs calibration points dense enough within each local region, an *infill asymptotic* requirement this AOI's validation split plausibly did not satisfy, a density limit, not evidence against localization. Nor is the pattern isolated to this paper: applying Mondrian calibration to real-estate valuation, Hjort

When node uncertainties were aggregated to 2.5 km pseudo-projects as VM0033 prescribes (root-sum-of-squares,

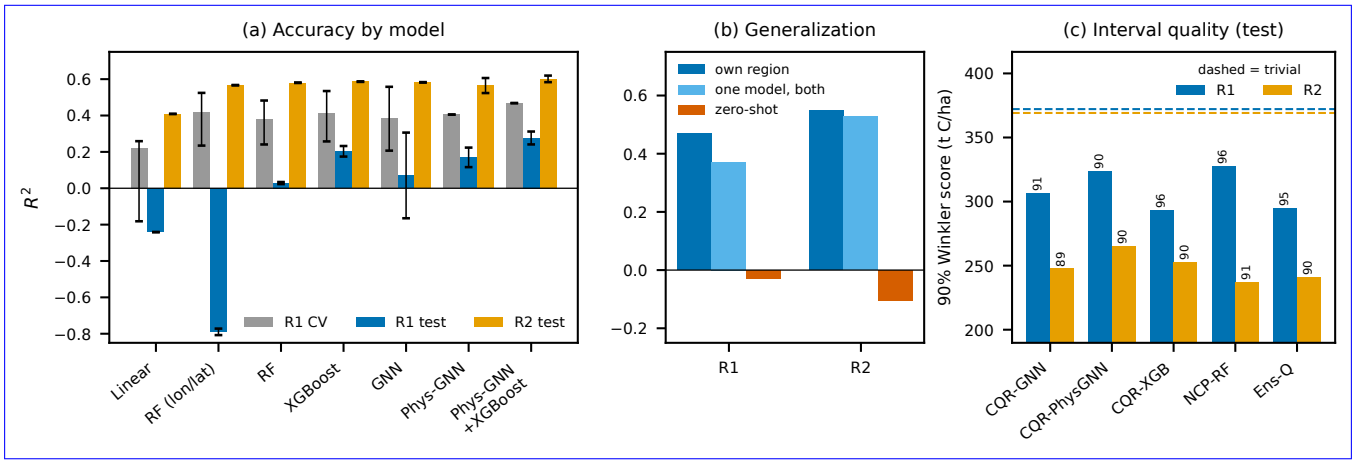


Fig. 1. (a) R^2 by model: R1 cross-validation (whiskers: fold range) and test in R1 and R2 (mean ± SD, 3 seeds). (b) Ensemble generalization: own-region and two-region cross-validation, and zero-shot transfer (XGBoost, test). (c) 90% Winkler score on test (lower is better); dashed lines: trivial interval; numbers: coverage (%).

TABLE II
INTERVALS (90%, TEST) AND CREDITING

	R1	R2
Coverage	91.5%	89.1%
Mean width / trivial (t C/ha)	273 / 274	191 / 282
Winkler score / trivial	306 / 372	247 / 369
ρ(width, abs. error)	0.54	0.15
Median VM0033 deduction per node	12.7%	2.1%
Project coverage, independent errors*	18%	33%
Project coverage, correlated errors*	91%	100%

*Realized errors of 2.5 km pseudo-projects, validation data.

independent errors), the stated project uncertainty covered the realized aggregate error in 18% (R1, 11 projects) and 33% (R2, 45 projects) of cases and implied no deduction. Treating errors as fully correlated covered 91% and 100% and implied median deductions of 14.6% and 3.1%.

D. Field soil cores

The 87 mangrove cores of the Coastal Carbon Library [14] were converted to 0–100 cm stocks, using a quadratic organic-matter conversion of the form in Craft et al. ~~[?] report "little to no improvement" over global in one scenario, cautioning that an apparent gain can instead reflect a non-conformity score failing to capture spatial trends. Localization gains in asset-pricing conformal calibration are setting-dependent, not automatic. global (simpler, the only method meeting nominal coverage on test) is defensible for near-term deployment; Section VI's pricing PoC uses global intervals accordingly. Both methods still produce test-set widths (318–383~~ [31] fitted on 679 regional samples ($R^2 = 0.91$), and matched to nodes within 300 ~~t/ha)several times the label's std. dev. (87.6~~m. At the 25 sites, the median absolute error was 10.1% for the ensemble, 11.6% for the physics-informed GNN, 16.5% for XGBoost, 14.1% and 15.1% for the Sanderman and Radabaugh maps, and 15.0% for a constant. The Sanderman map ranked sites best (Spearman 0.63; ensemble 0.29). The two published maps were uncorrelated at node level (−0.10 in R1).

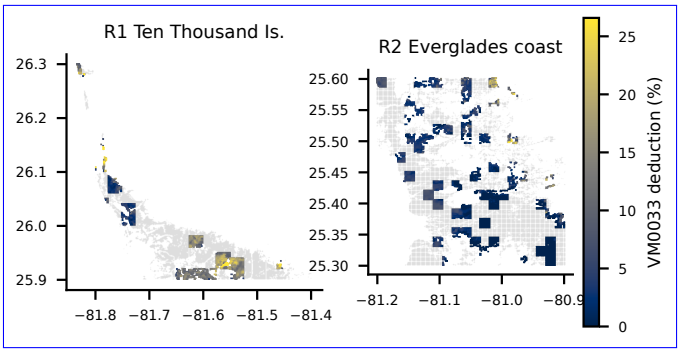


Fig. 2. Per-node VM0033 uncertainty deduction (Eq. 2) on test nodes of both regions, from 90% conformal intervals. Gray: training and validation nodes.

For the 13 cores sampled to at least 50 ~~t/ha), wide, not falsely confident~~cm, field-calibrated intervals covered 85–92% under leave-one-study-out and implied VM0033 deductions of 22–29%.

~~CQR interval width measures present-day estimation confidence,~~

V. DISCUSSION

The physics-informed GNN and tree ensembles are complementary. Alone, the physics-informed GNN matched XGBoost in cross-validation (0.41 each), but their mean improved both regions (0.47 and 0.55) and held on test (0.28 and 0.60). The GNN contributes neighborhood structure and the hydroperiod prior that the trees do not see, consistent with the control of tidal setting on mangrove soil carbon [18], and it was also the most stable graph model across seeds and, once trained on both dates, across seasons. The physics prior is now a weak constraint (correlation 0.14 with the predictions), not ~~non-permanence risk (future carbon loss to erosion, disturbance, sea-level rise) – the quantity buffer pools price; this paper does not establish the two correlate. We illustrate, without validating, how~~ a target the model copies.

Estimation uncertainty is as large as the buffer the market already holds. The per-node ~~estimation confidence~~

could sit alongside, not replace, a non-permanence buffer, built on *global* CQR intervals per Section V's recommendation, not deduction in R1 (12.7%) is similar to the realized median buffer of issuing blue carbon projects (11.0%), yet today it is not applied. Fig. 2 shows that the deduction concentrates in specific areas, so it is a location-dependent quantity that a flat buffer cannot represent. Because width does not follow observed mangrove loss, the localized (Mondrian) variant it cautions against. A relative interval width is the 90%-CQR (global) width divided by the test set's own median (341.1 t/ha; range 143.9–552.5 t/ha). A price multiplier is $1 - \alpha$(relative width $- 1$), clipped to $[0.5, 1.5]$, with $\alpha = 1.0$ an arbitrary illustrative sensitivity, not fit to market outcome. Table **??** shows five nodes against Plan Vivo V5.0's flat 20% buffer . two discounts measure different risks and belong in separate factors, as in Eq. 2. **Business implication:** a registry can adopt the deduction without changing its buffer policy.

Aggregation under independent errors is the weak point of the registry rule. Spatial model errors are correlated, so root-sum-of-squares aggregation shrinks project uncertainty and grants no deduction even where realized errors exceed the allowance. A correlation-aware or empirically calibrated aggregation restored nominal coverage. The same dependence explains why random validation overstates mapping accuracy [22] and why Mondrian calibration undercovered with only six blocks [25]. Model-based crediting therefore needs an aggregation rule that accounts for spatial dependence.

Concretely, an issuer selling from both the well- (min-width) and poorly-characterized (max-width) node above applies the identical 20% discount today; under a confidence-adjusted mechanism, the former prices at a 50% premium, the latter at a 50% discount, a plausible implication, not a validated outcome. α and the clipping bounds are stated, arbitrary choices. The flat-buffer value is not arbitrary: it is Plan Vivo V5.0's stated buffer, used only as a comparison point (VM0033 instead scores a project-specific value via its own risk tool, Section I). Fig. **??** maps this interval width across the full test set, the spatial pattern behind the location-differentiated pricing Table **??** illustrates. **What the intervals can and cannot do.** Interval width is bounded by accuracy: with test $R^2 = 0.07$ for the GNN underlying the intervals in R1, at most about 4% narrowing over the trivial interval is attainable, and in R2 ($R^2 = 0.58$) the 32% achieved is close to the 35% bound. The value in R1 lies in adaptivity and Winkler score, not sharpness. Local calibration is required when a model is moved to a new region, and at core scale even field-calibrated maps barely beat a regional mean, while at site scale the ensemble did. Field-calibrated deductions (22–29%) exceed label-calibrated ones, a reminder that the label product carries its own error, as fine-scale disagreement between soil carbon maps elsewhere has shown [20].

VI. Limitations

Labels are a modeled product: the training label is a global map that disagrees with a field-calibrated regional map at node level. **Accuracy in R1 is moderate** ($R^2 = 0.28$). **The ensemble was evaluated in a second test read,** declared and limited to one pre-specified model. **Single tide gauges** represent each region; R2's is about 70 km away. **Field cores are sparse** (13 deep enough for 0–100 cm stocks) and may overlap the global map's training data. **Market outcomes** were not modeled; prices and buffers come from published records.

constrains the physics term to a cross-sectional relationship, not an actual dC/dt (not invalidating the prior-generalization result, defined relative to that same proxy). exists for this AOI; accretion/export are proxies (Section III-E). , not a tidal-energy layer. (Sections III-F, V): it measures transfer, not physical correctness. , a disclosed scope choice. (Section IV-C, V) , a data-density limit here, not spatial calibration generally [6], [25]. ; the five-seed check addresses seed variance, not regional generalization. , with an unfit sensitivity parameter and no market validation. (9.7% of nodes, 14.2% of test nodes, are 1-hop neighbors across a split); restricting Table **??** to the 1,782 non-boundary test nodes gives lower R^2 for all three models (linear -0.504, baseline GNN -0.447, physics GNN -0.494) but the same ranking, so reported numbers carry a small, non-model-specific optimistic bias from this leakage. remain unaddressed.

VII. Conclusion

A pre-registered, two-stage protocol (validation-only selection, single held-out test confirmation) caught two overstated results a single run would have missed: an earlier single-seed pilot's headline finding, and a theoretically-motivated localized calibration refinement favorable on validation-internal evidence alone. What survives (the Calibrated per-node uncertainty from a physics-informed model's smaller but real prior-generalization advantage, GNN and XGBoost ensemble can supply the VM0033 uncertainty deduction. Across two regions and two image dates, the ensemble reached test $R^2 = 0.28$ and 0.60, intervals reached 89–92% coverage, and the deduction (median 12.7% in R1) is comparable to the 11.0% buffers that issuing projects hold. The methodology's independent-error aggregation covered only 18–33% of realized project errors, and against 87 field cores the ensemble's site-level error (10.1%) was below both published maps and a regional mean. Future work will test a correlation-aware aggregation with registry project data, add regions with denser field cores, and honestly wide, node-differentiated CQR intervals) are two separate results, not one flowing from the other: the CQR pipeline priced here used data loss only , without the physics term. Together they motivate the market-facing contribution: an estimation-confidence signal that could complement, not replace, flat buffer pools, deployed here on the simpler, better-calibrated global variant. Future work : a second Sentinel-2 date, a higher-volume or second-region localized-conformal test, deep-ensemble/MC-dropout comparisons, and market-outcome validation of the pricing mechanism against non-permanence riskreplace

the steady-state prior with repeated elevation and accretion measurements.

Code Availability

Code, data pipeline, selection rules and results: https://github.com/Nahla-Nabil/PIGNN-UQ.

References

[1] Verra, "VM0033: Methodology for tidal wetland and seagrass restoration, v2.1," https://verra.org/methodologies/vm0033-methodology-for-tidal-wetland-and-seagrass-restoration-v2-1/, 2023.

[2] Plan Vivo Foundation, "Non-permanence buffer, Plan Vivo standard V5.0," https://www.planvivo.org/buffer, 2022.

[3] K. R. Radabaugh *et al.*, "A spatial model comparing above- and belowground blue carbon stocks in southwest Florida mangroves and salt marshes," *Estuaries and Coasts*, vol. 46, pp. 1536–1556, 2023, doi: 10.1007/s12237-023-01217-7.

[4] J. Sanderman *et al.*, "Global mangrove soil organic carbon stocks dataset at 30 m resolution for the year 2020 based on spatiotemporal predictive machine learning," *Data in Brief*, vol. 50, p. 109621, 2023, doi: 10.1016/j.dib.2023.109621.

[5] G. Singh, G. Moncrieff, Z. Venter, K. Cawse-Nicholson, J. Slingsby, and T. B. Robinson, "Uncertainty quantification for probabilistic machine learning in earth observation using conformal prediction," *Scientific Reports*, vol. 14, p. 16166, 2024, doi: 10.1038/s41598-024-65954-w.

[6] X. Lou, P. Luo, and L. Meng, "GeoConformal prediction: A model-agnostic framework for measuring the uncertainty of spatial prediction," *Annals of the American Association of Geographers*, vol. 115, no. 8, pp. 1971–1998, 2025, arXiv:2412.08661.

[7] M. Raissi, P. Perdikaris, and G. E. Karniadakis, "Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations," *Journal of Computational Physics*, vol. 378, pp. 686–707, 2019, doi: 10.1016/j.jcp.2018.10.045.

[8] M. Taghizadeh, Z. Zandsalimi, M. A. Nabian, M. Shafiee-Jood, and N. Alemazkoor, "Interpretable physics-informed graph neural networks for flood forecasting," *Computer-Aided Civil and Infrastructure Engineering*, vol. 40, pp. 2629–2649, 2025, doi: 10.1111/mice.13484.

[9] W. L. Hamilton, R. Ying, and J. Leskovec, "Inductive representation learning on large graphs," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2017, arXiv:1706.02216.

[10] L. Breiman, "Random forests," *Machine Learning*, vol. 45, pp. 5–32, 2001, doi: 10.1023/A:1010933404324.

[11] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2016, pp. 785–794, doi: 10.1145/2939672.2939785.

[12] Y. Romano, E. Patterson, and E. J. Candès, "Conformalized quantile regression," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2019, arXiv:1905.03222.

[13] P. Quartson *et al.*, "Voluntary Registry Offsets Database v2026-08," https://gspp.berkeley.edu/berkeley-carbon-trading-project/offsets-database, 2026, Berkeley Carbon Trading Project, University of California, Berkeley.

[14] Coastal Carbon Network, "Database: Coastal Carbon Library (version 1.7.0)," Smithsonian Environmental Research Center, dataset, 2023, doi: 10.25573/serc.21565671, accessed 2026-10-09.

[15] J. Willard, X. Jia, S. Xu, M. Steinbach, and V. Kumar, "Integrating scientific knowledge with machine learning for engineering and environmental systems," *ACM Computing Surveys*, vol. 55, no. 4, pp. 1–37, 2023, doi: 10.1145/3514228.

[16] A. D. Roy *et al.*, "Remote sensing-based mangrove blue carbon assessment in the Asia-Pacific: A systematic review," *Science of The Total Environment*, vol. 938, p. 173270, 2024, doi: 10.1016/j.scitotenv.2024.173270.

[17] J. Sanderman *et al.*, "A global map of mangrove forest soil carbon at 30 m spatial resolution," *Environmental Research Letters*, vol. 13, p. 055002, 2018, doi: 10.1088/1748-9326/aabe1c.

[18] A. S. Rovai *et al.*, "Global controls on carbon storage in mangrove soils," *Nature Climate Change*, vol. 8, pp. 534–538, 2018, doi: 10.1038/s41558-018-0162-5.

[19] K. W. Krauss *et al.*, "How mangrove forests adjust to rising sea level," *New Phytologist*, vol. 202, pp. 19–34, 2014, doi: 10.1111/nph.12605.

[20] J. R. Holmquist *et al.*, "Accuracy and precision of tidal wetland soil carbon mapping in the conterminous United States," *Scientific Reports*, vol. 8, p. 9478, 2018, doi: 10.1038/s41598-018-26948-7.

[21] D. R. Roberts *et al.*, "Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure," *Ecography*, vol. 40, pp. 913–929, 2017, doi: 10.1111/ecog.02881.

[22] P. Ploton *et al.*, "Spatial validation reveals poor predictive performance of large-scale ecological mapping models," *Nature Communications*, vol. 11, p. 4540, 2020, doi: 10.1038/s41467-020-18321-y.

[23] J. Lei, M. G'Sell, A. Rinaldo, R. J. Tibshirani, and L. Wasserman, "Distribution-free predictive inference for regression," *Journal of the American Statistical Association*, vol. 113, pp. 1094–1111, 2018, doi: 10.1080/01621459.2017.1307116.

[24] V. Vovk, A. Gammerman, and G. Shafer, *Algorithmic Learning in a Random World*. Springer, 2005, doi: 10.1007/b106715.

[25] H. Mao, R. Martin, and B. J. Reich, "Valid model-free spatial prediction," *Journal of the American Statistical Association*, vol. 119, no. 546, pp. 904–914, 2024, doi: 10.1080/01621459.2022.2147531.

[26] R. L. Winkler, "A decision-theoretic approach to interval estimation," *Journal of the American Statistical Association*, vol. 67, pp. 187–191, 1972, doi: 10.1080/01621459.1972.10481224.

[27] T. Gneiting and A. E. Raftery, "Strictly proper scoring rules, prediction, and estimation," *Journal of the American Statistical Association*, vol. 102, pp. 359–378, 2007, doi: 10.1198/016214506000001437.

[28] S. Farahmand, N. Hilmi, and C. M. Duarte, "The rise and flows of blue carbon credits advance global climate and biodiversity goals," *npj Ocean Sustainability*, vol. 4, p. 39, 2025, doi: 10.1038/s44183-025-00141-6.

[29] P. Bunting *et al.*, "Global mangrove extent change 1996–2020: Global Mangrove Watch version 3.0," *Remote Sensing*, vol. 14, p. 3657, 2022, doi: 10.3390/rs14153657.

[30] M. Drusch *et al.*, "Sentinel-2: ESA's optical high-resolution mission for GMES operational services," *Remote Sensing of Environment*, vol. 120, pp. 25–36, 2012, doi: 10.1016/j.rse.2011.11.026.

[31] C. B. Craft, E. D. Seneca, and S. W. Broome, "Loss on ignition and Kjeldahl digestion for estimating organic carbon and total nitrogen in estuarine marsh soils: Calibration with dry combustion," *Estuaries*, vol. 14, pp. 175–179, 1991, doi: 10.2307/1351691.